



Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders with Social Distancing

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By [Chandra, Shailesh](#) ; [Mishra, Vivek](#)

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Creators:

Chandra, Shailesh ; Mishra, Vivek

Corporate Creators:

Mineta Transportation Institute ; San Jose State University. College of Business

Corporate Contributors:

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Abstract:

During the COVID-19 pandemic, LA Metro has encouraged social distancing among passengers—especially at stations of high-demand routes—and has increased fixed-route transit (FRT) services. However, potential impacts of social distancing on the performance of FRT services remain mostly unknown. This research evaluates the accessibility of FRT buses with social distancing using the ridership data collected on four FRT routes: 105, 108, 111, and 115 of the LA Metro's A Line stations located in low-income neighborhoods. This research shows that social distancing of six feet can impact FRT's accessibility to destination stations, and maximum accessibility is achieved only for a certain number of stops served—which is less than the current number of stops served. The FRT routes 105, 108, 111 and 115 have maximum accessibility with social distancing for the number of stops served equal to 65, 52, 52 and 50, respectively. The methodology used in this research can help decision-makers understand how FRT bus frequencies are impacted by social distancing measures, and the results can guide the transit authorities developing FRT service among low-income commuters during and after the pandemic.

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